

Chapter – 11: Light: Shadows and Reflections

- Western Ghat region – Maharashtra – Keshav – spends vacation – friend Jatin’s Grandparents’ village
- Lived – big city – forests, fresh air, gushing streams, chirping birds – novel experience
- Keshav – most fascinating – fireflies – night – wonderful experience
- Grandparents – explain – fireflies – seasonal insects – use light – communicate
- Keshav – also learns – no. of fireflies – decreasing – light pollution, reduced cover, excessive tourism
- Vacation end – Keshav, Jatin – board – evening bus – return to city
- Bus – travel – hilly roads – Keshav watch – moonlit landscape – beams of headlights
- Remembers – poems, songs – moonlight – wonders – moon – own light?
- Grade 6 – *curiosity* – chapter – beyond earth – learnt – all obj. – solar sys. – reflect light – sun
- Also notice – strange – light – travel – straight line

Sources of Light

- Sun – give out, emit – own light – main src. – natural light – Earth
- Stars, lightning, natural fire, certain animals – emit – own light
- Ancient times – humans – learnt – create fire – earliest form – artificial lighting
- With time – learnt – create fire – diff. fuels – animal fat, oil, wax, gas
- Invention – lighting – diff. elec. src. – most needs – met by – elec. lighting
- Obj. – emit own light – **luminous**
- Obj. – do not emit own light – **non-luminous**
- Moon – non-luminous – no own light – reflects – sun’s light

Does Light Travel in Straight Line?

- Activity –
 - Take – 3 matchboxes – make hole – each inner tray – same position
 - Arrange – matchbox – straight line – make sure – holes – straight line
 - Place – torch – one side – matchbox – ensure – lamp – lines up – holes
 - Place – cardboard (screen) – other side – obt. – bright spot
 - Move – 1 matchbox – side, up, down – obt. – light spot?
- All 3 holes – no straight line – cannot obt. – light spot
- Activity –
 - Take – long hollow pipe – flexible material – align – view – candle flame
 - Bend pipe – try to view – candle flame – not visible
- Candle flame – visible – straight pipe – not visible – bent pipe – proves – light – travel – straight line

Light through Transparent, Translucent, Opaque Materials

- Activity –
 - Collect – torch, obj. – diff. materials – make list
 - **Classify** – transparent, translucent, opaque – use knowledge – grade 6 – *curiosity* – chapter – materials around us
 - Go – dark room – turn on torch – place it – position – light spot – wall
 - Conduct act. – 2 parts – prediction, observation –

- **Predict** – hold obj. – light front – continue vision – light spot?
- Place obj. – b/w – torch, screen – note obs.
- Repeat – all obj.
- Obs. – same – prediction? – what conclusions?
- Light – pass completely – **transparent** – partially – **translucent** – does not pass – **opaque**

Shadow Formation

- Above activity – opaque obj. – light path – dark patch – visible on wall
- We know – light travels – straight line
- Opaque obj. – placed in path – light blocked
- Dark patch – no light – **shadow**
- Notice – shadows – ourselves, other obj. – Sun, under light
- Have fun – make diff. shapes – shadows
- Translucent, transparent obj. – create shadows or not?
- **Notice** – opaque obj. – darker shadows – translucent – lighter shadows – transparent – faint shadows
- Activity –
 - Collect – opaque obj. – diff. shapes, sizes
 - Repeat activity – above – this time – fiddle with – position – screen, light src. – obj. – colour, shape, size
 - **Observe** – shadow – shape, size – each action
 - Shape, size – same – each case?
 - **Record** – obs.
- **Conclusions** – above act. – obs. shadow – req. – any colour change – change of obj.?
- Shadows formed – obj. – blocks light – falling on screen
- Req. – light src., opaque obj., screen – obs. shadow
- Walls, floor, grnd., any surface – screen – daily life
- Shape, size, sharpness – shadow – depend – position – obj. – rel. – light src., screen
- Shadows – give some info. – OR – not be able to guess
- Changing colour – obj. – no colour change – shadows

Reflection of Light

- Opaque obj. – shiny obj. – polished steel plate – shadow – screen – BUT also – bright light spot – wall – opposite side
- Activity –
 - Find – shiny steel plate, plane mirror – not curved
 - Take – outside – sunlight – fall – shiny surface
 - Turn – plate, mirror – diff. dir. – redirect light – wall, nearby surface
 - Tilt – plate, mirror – diff. ways – obs. – light spot
 - Notice – light – travel – st. line – change dir. – falls – shiny surface
- This activity – suggest – shiny surface, mirror – change – dir. of light
- This change – dir. of light – mirror – **reflection of light**
- Activity –
 - Take – plane mirror, stand, torch, comb, white paper sheet, black paper strip
 - Use black paper – cover comb – leave 1 opening – make thin slit

- Spread – white paper – table – hold comb – perpendicular – shine torch – slit – adjust – comb, torch – thin beam obtained
- Place – mirror – light path – keep comb steady – obs.
- Light path – change – falling on mirror – reflection – occurs at mirror

Images Formed in Plane Mirror

- Look into mirror – view – your face – your reflection
- Activity –
 - Take – plane mirror, pen, other obj.
 - Place – pen – front of mirror
- Obs. – similar pen – visible – behind mirror – **image** – pen – **object**
- Move – pen – diff. pos. – **compare** – img. sizes – pen – each pos.
- Img. – plane mirror – same size – obj.
- Move – pen – diff. pos. – obs. – img. – upright – each pos.
- Upright img. – **erect** – plane mirror – always erect
- Place screen – behind mirror – move around – any img.? – repeat – screen – mirror front
- Img. – plane mirror – cannot be obt. – screen
- Activity –
 - Stand – front of mirror – look at img. – notice – dist. from mirror
 - Stand closer – img. – also closer?
 - Stand – diff. dist. – notice – dist. – img. – each case
- Notice – stand close – plane mirror – img. – closer – stand far – img. also far
- Raise – left arm – which arm – img. raise?
- Touch – right ear – which ear – img. touch?
- Notice – your left – appear right – your right – appear left
- This type – perceived left-right reversal – **lateral inversion**
- AMBULANCE – written – reversed form – viewed straight – rear-view mirror

Pinhole Camera

- Pinhole camera – device – light rays – obj. – pass through – tiny hole – form img. – screen
- Activity –
 - Take – cardboard, candle – make small hole – cardboard
 - Dimly lit room – position – cardboard – short dist. – screen
 - Place – lighted candle – in front
- Obs. – screen – light from candle – pass through cardboard hole – forms img. – screen
- Img. – candle flame – upside down – inverted
- Activity –
 - Take – 2 boxes – 1 slides into another – little gap – cut open – 1 side – each box
 - Opp. face – larger box – make hole – small
 - Opp. face – smaller box – cut – sq. hole – middle – 5-6 cm – cover opening – thin translucent paper – make screen
 - Slide – smaller box – inside larger box – side – tracing paper – inside
- Hold – pinhole camera – pinhole – face obj. – look through – open side – smaller box
- Cover – head, camera – dark cloth

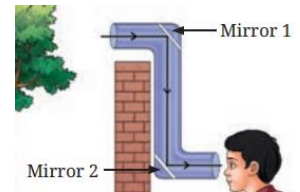
- Look – distant obj. – tree, building – move – smaller box – img. obt. – screen
- Img. – no colour – inverted

Making Some Useful Items

- Learnt – light – travel – st. line – reflected – mirrors – its time – **create** – useful items

Periscope

- 2 plane mirrors – face each other – each bend – Z-shape box
- Reflection – 2 mirrors – enables – view obj. – not visible directly
- Periscopes – used – submarines, tanks, bunkers



Kaleidoscope

- 3 rectangular plane mirror strips – equal size – join them – triangular manner
- Fix them – circular tube – thick chart paper – rubber band, adhesive tape
- Tube – 1 end – fix – transparent plastic sheet
- Place – several broken pieces – coloured bangles – cover it – tracing paper
- Peep through – open side – beautiful patterns
- Interesting pattern – kaleidoscope – always – diff. patterns
- 3 mirrors – multiple img. – multiple reflections – many interesting patterns

